Contents

1. OBJECTIVES:

The aim of this experiment is to help students to find out some of the basic theories of the studied theory in the field of thermodynamics.

It helps students have a general concept of the subject, their role and application in industry and life.

2. EXPERIMENTAL THEORY

2.1. Classification of moist air condition

◆ *Unsaturated moist air* : the type of moist air that the amount of vapor contains in which isn't maximum. Unsaturated moist air has ability to contain more water. The condition of vapor in unsaturated moist air is superheated steam. Vapor pressure of unsaturated moist air is smaller than the saturated pressure corresponding to moist air temperature. ($P_h < P_{hs}$).

◆ *Saturated moist air* : the type of moist air that the amount of vapor contains in which is maximum $G_h = G_{hmax}$. The condition of vapor in saturated moist air is dry saturated steam. Vapor pressure of saturated moist air is equal to the saturated pressure corresponding to moist air temperature. ($P_h = P_{hs}$)

◆ *Supersaturated moist air:* the type of moist air that the amount of vapor contains in which is maximum and can contain more water condensate. If the temperature is lower than 0°C, there will be ice. The condition of vapor in supersaturated moist air is moist saturated steam.

2.2. Characteristic parameters of moist air

PARAMETER	SYMBOL	UNIT	DEFINITION
Relative humidity	φ	%	- The ratio (expressed as a percentage) of the water vapor in the air at a certain temperature and pressure to the maximum water vapor it can contain at the same temperature and pressure. $\phi = \frac{P_h}{P_{bh}} 100\% = \frac{\rho_h}{\rho_{bh}} 100\%$ P_h, P_{bh} : vapor partial pressure and vapor saturated pressure of water at the same temperature condition
Absolute humidity	x (d,y)	kg_{h20}/kg_{dry_air}	- Density of moisture (water vapor) per unit volume of air, expressed usually as kilogram per cubic meter (kg/m3)). $x = \frac{M_h}{M_{kkk}} \times \frac{P_h}{P - P_h} = \frac{18}{29} \times \frac{\phi P_{bh}}{P - \phi P_{bh}}$
Heat function	H (I)	KJ/ kg dry air	**$H = C_{kkk}.t + (r + C_h t).x = t + (2493 + 1,97t).x$** C_{kkk} = 1 (kJ/kg. °C): specific heat of dry air t (°C) : temperature of dry air r = 2493 (kJ/kg. °C): latent heat of water at 0°C C_h = 1,97 (kJ/kg. °C): specific heat of

			steam
Dry bulb temperature	t (t_{dry}, τ)	°C	- Refers basically to the ambient air temperature. It is called "Dry bulb" because the air temperature is indicated by a thermometer not affected by the moisture of the air.
Wet bulb temperature	t_{wet}	°C	- Wet bulb temperature can be measured by using a thermometer with the bulb wrapped in wet muslin. The adiabatic evaporation of water from the thermometer bulb and the cooling effect is indicated by a "wet bulb temperature".
Drying potential	ε	°C	- Difference between dry bulb temperature and wet bulb temperature $\boldsymbol{\varepsilon = t_{dry} - t_{wet}}$
Dew point	T_{dp}	°C	- The dew point temperature t_r is defined as the temperature at which condensation begins if the air is cooled at constant pressure.

2.3. Classification of steam condition

- ◆ *<u>Saturated steam</u>*: When the water evaporated at boiling temperature
- ◆ *<u>Superheated steam</u>:* is saturated steam heated up without changing vapor pressure

3. EXPERIMENTAL EQUIPMENT:

3.1. Experimental model :

The principle diagram of the experimental model is shown in Figure 1. It is an aerodynamic tube. In which the air is passed from one end of the tube to the other and the temperature is cooled by evaporator of air conditioner, dried by resistor, moistened by spraying vaporized water from the vaporizer.

3.2. Diagram description:

Air through fan (with flowrate adjustment) 1 passes through aerodynamic tube 2 that is cooled in cooler 4. Then heated and dried in dryer 5, after that, air is moistened by steam nozzle 6 and goes out. In the front and behind positions of each equipment in the aerodynamic tube, there are dry bulbs 7 and wet bulb 8 to measure the temperature and humidity of the air. At the outside aerodynamic tube, there is a speedometer 9 to determine the flowrate of air. Below colder 4, it is equipped with a volume measuring device to determine the flowrate of the condensed water from cold air.

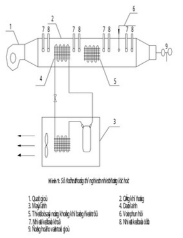

Figure 1 : Experimental diagram of Thermodynamics

1. Fan
2. Aerodynamic tube
3. Air conditioner
4. Cooler
5. Drying equipment by resistor
6. Steam nozzle
7. Dry bulb thermometer
8. Wet bulb thermometer
9. Speedometer

4. EXPERIMENTAL METHOD :

In this experiment, students must carry out the following tasks:

1. Determine the status of air that includes: temperature, humidity of the cooler 4(the status of air temperature in the surrounding environment), before the air drying 5 (behind the cooler 4), before the steam nozzle 6 and after the steam nozzle *exhale). From the measured data, the students must draw changing status of air on I-D diagram and on that basic students must determine the enthalpy and the vapor density of the locations mentioned above.
2. Calculate the heat balance of the aerodynamic tube including: determining the flowrate of air, determining the cold yield of the cooler and the heating load of the drying equipment.

◆ ***<u>Operation process:</u>***

1. Turn on the gate switch, check the 3-phase indicator on the main switchboard.
2. Turn on fan, air flowrate adjustment by opening/closing the door.
3. Turn on air conditioner.
4. Turn on resistor.
5. Turn on the steam control for saturated steam. Observe temperature and pressure in the steam tank, if pressure is reached 2 kg/cm^2, start the steam valve.
6. After open the steam valve, let the system run for 2 minutes to achieve stability.

In turn measure dry and wet bulb temperatures at locations.
Use a measuring tube and a meter to measure the condensed water flow after the cooler.

7. Turn on the steam control for superheated steam, let the system run for 2 minutes and do measurement like above.
8. Change the other operating mode by changing the wind outlet position, increasing or decreasing the resistor, increasing or decreasing the airflow.

Note:

The water level in the boiler was tested after each experiment (turn off the resistor) by opening/closing the valve between the boiler and the water tank to provide water for the boiler. Provided water level is equal to saturated steam thermometer.

5. RESULT:

Table 1: Temperature of air

Saturated steam

Velocity of wind at outlet of dynamic gas pipe v (m/s)	No.	Point 1		Point 2		Point 3		Point 4	
		Front of cooler (environment)		Front of drying (behind cooler)		Front of steam nozzel (behind drying)		Behind steam nozzel (eliminate to environment)	
		Dry bulb temp.	Wet bulb temp.	Dry bulb temp.	Wet bulb temp.	Dry bulb temp.	Wet bulb temp.	Dry bulb temp.	Wet bulb temp.
2.6	1	30.10	26.40	23.00	21.60	44.80	30.90	34.00	32.70
	2	30.40	26.50	18.40	18.30	43.40	29.30	32.90	32.00
	Ave	**30.25**	**26.45**	**20.70**	**19.95**	**44.10**	**30.10**	**33.45**	**32.35**
3.5	1	30.40	26.30	15.40	14.30	37.80	26.90	31.00	29.90
	2	30.40	26.40	15.60	15.40	37.50	26.70	31.50	30.50
	Ave	**30.40**	**26.35**	**15.50**	**14.85**	**37.65**	**26.80**	**31.25**	**30.20**
4.1	1	30.40	26.30	24.00	19.80	39.70	28.90	32.70	30.50
	2	30.70	26.40	25.90	19.60	40.50	29.30	32.90	30.70
	Ave	**30.55**	**26.35**	**24.95**	**19.70**	**40.10**	**29.10**	**32.80**	**30.60**

Superheated steam

Velocity of wind at outlet of dynamic gas pipe v (m/s)	No.	Point 1		Point 2		Point 3		Point 4	
		Front of cooler (environment)		Front of drying (behind cooler)		Front of steam nozzel (behind drying)		Behind steam nozzel (eliminate to environment)	
		Dry bulb temp.	Wet bulb temp.	Dry bulb temp.	Wet bulb temp.	Dry bulb temp.	Wet bulb temp.	Dry bulb temp.	Wet bulb temp.
2.6	1	30.70	26.50	20.50	19.20	44.50	29.80	35.20	33.00
	2	30.30	25.90	18.10	17.90	42.60	28.70	34.90	31.80
	Ave	**30.50**	**26.20**	**19.30**	**18.55**	**43.55**	**29.25**	**35.05**	**32.40**
3.5	1	30.40	26.10	22.90	18.30	40.60	28.90	34.00	31.20
	2	30.40	26.10	20.20	18.60	39.20	28.30	33.60	31.30
	Ave	**30.40**	**26.10**	**21.55**	**18.45**	**39.90**	**28.60**	**33.80**	**31.25**

4.1	1	30.50	26.10	22.90	19.60	39.70	28.70	33.70	31.70
	2	30.50	26.10	24.40	19.70	39.90	28.80	33.30	30.10
	Ave	**30.50**	**26.10**	**23.65**	**19.65**	**39.80**	**28.75**	**33.50**	**30.90**

Table 2: Other data

Saturated steam

Velocity of wind at outlet of aerodynamic tube *v* (m/s)	No.	Water sample from cooler V1 (ml)	Time to take water below t1 (s)
2.6	1^{st}	16	60
	2^{nd}	20	60
3.5	1^{st}	15	60
	2^{nd}	17	60
4.1	1^{st}	7.3	60
	2^{nd}	7.3	60

Superheated steam

Velocity of wind at outlet of aerodynamic tube *v* (m/s)	No.	Water sample from cooler V1 (ml)	Time to take water below t1 (s)
2.6	1^{st}	8.7	60
	2^{nd}	8.7	60
3.5	1^{st}	7.3	60
	2^{nd}	6.7	60
4.1	1^{st}	6	60
	2^{nd}	4.5	60

Table 3: Parameters of humid air

Saturated steam

Velocity of wind at outlet of dynamic gas pipe v (m/s)	Parameter	Temperature t (^{O}C)		Relative humidity ϕ (%)	Enthalpy I (kJ/kg)	Density d (kg/kg)
		Dry bulb temp. (^{O}C)	Wet bulb temp. (^{O}C)			
2.6	**Point 1**	30.25	26.45	74.4	82.6	0.0204
	Point 2	20.70	19.95	93.5	57.4	0.0144
	Point 3	44.10	30.10	37.1	99.9	0.0216
	Point 4	33.45	32.35	92.6	113.0	0.0310
3.5	**Point 1**	30.40	26.35	73.0	82.2	0.0202
	Point 2	15.50	14.85	93.7	41.7	0.0103
	Point 3	37.65	26.80	43.3	83.8	0.0179
	Point 4	31.25	30.20	92.7	101.0	0.0272
4.1	**Point 1**	30.55	26.35	72.1	82.1	0.0201
	Point 2	24.95	19.70	61.8	56.4	0.0123
	Point 3	40.10	29.10	44.7	94.8	0.0212

	Point 4	32.80	30.60	85.4	103.1	0.0274

Superheated steam

Velocity of wind at outlet of dynamic gas pipe v (m/s)	**Parameter**	**Temperature t (^{O}C)**		**Relative humidity ϕ (%)**	**Enthalpy I (kJ/kg)**	**Density d (kg/kg)**
		Dry bulb temp. (^{O}C)	**Wet bulb temp. (^{O}C)**			
2.6	**Point 1**	30.50	26.20	71.5	81.5	0.0199
	Point 2	19.30	18.55	93.2	52.7	0.0131
	Point 3	43.55	29.25	35.3	95.4	0.0200
	Point 4	35.05	32.40	83.2	113.3	0.0304
3.5	**Point 1**	30.40	26.10	71.4	81	0.0197
	Point 2	21.55	18.45	74.7	52.3	0.0121
	Point 3	39.90	28.60	43.4	92.3	0.0203
	Point 4	33.80	31.25	83.5	106.6	0.0283
4.1	**Point 1**	30.50	26.10	70.9	81	0.0197
	Point 2	23.65	19.65	69.4	56.2	0.0128
	Point 3	39.80	28.75	44.3	93.1	0.0206
	Point 4	39.80	28.75	83.1	104.7	0.0277

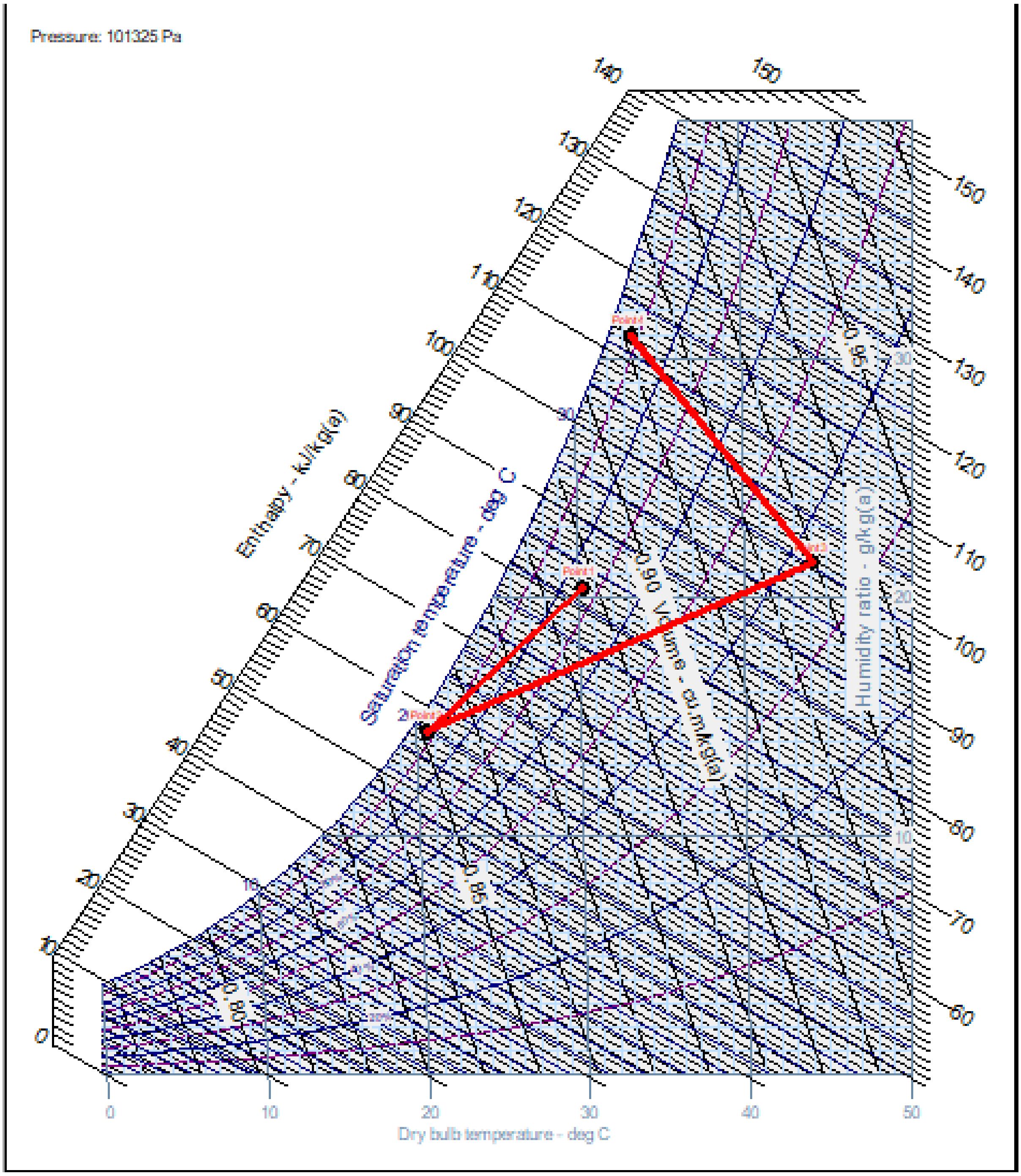

Figure 2: The process of the changing of airflow at v = 2.6 m/s for saturated steam

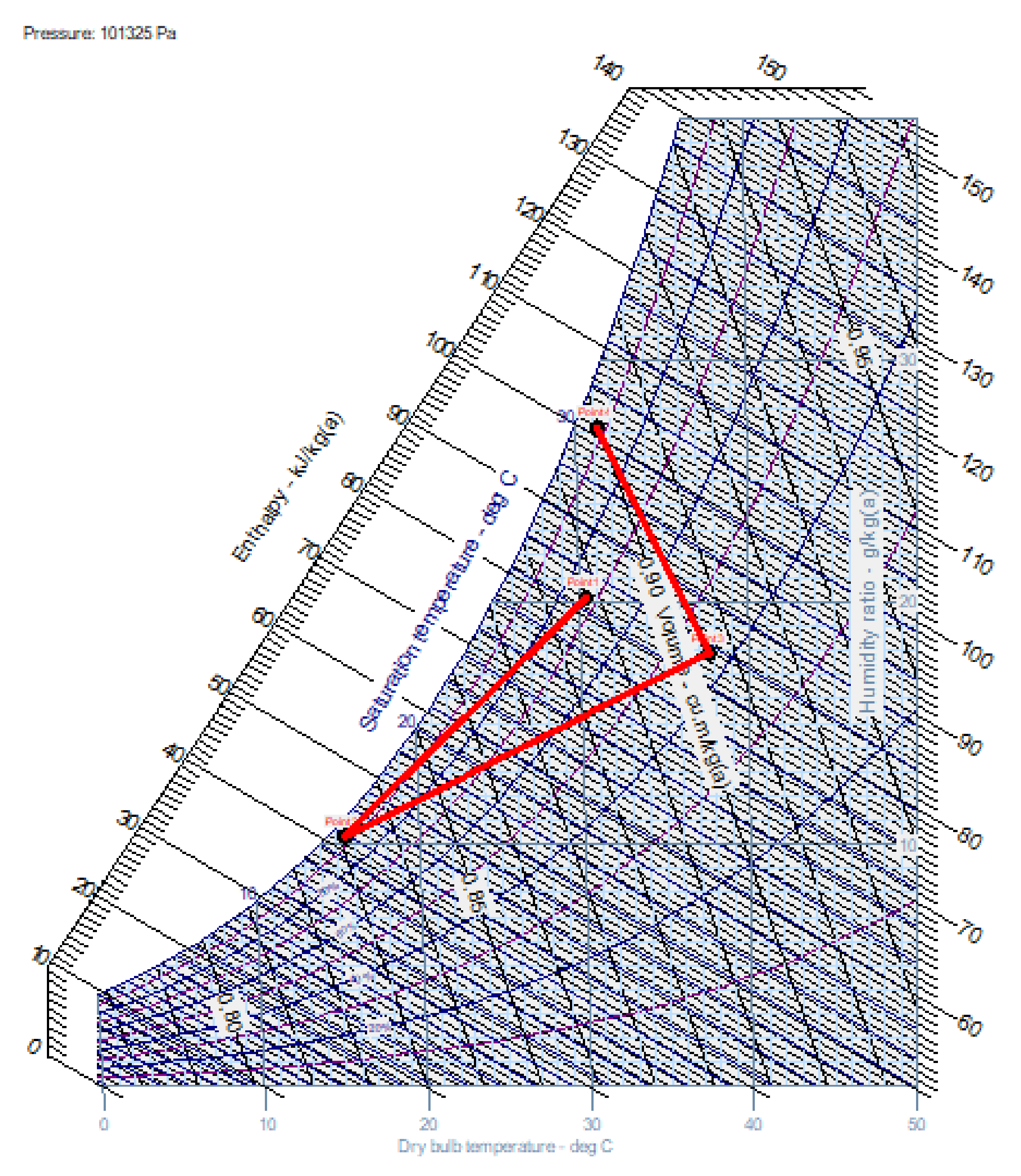

Figure 3: The process of the changing of airflow at v = 3.6 m/s for saturated steam

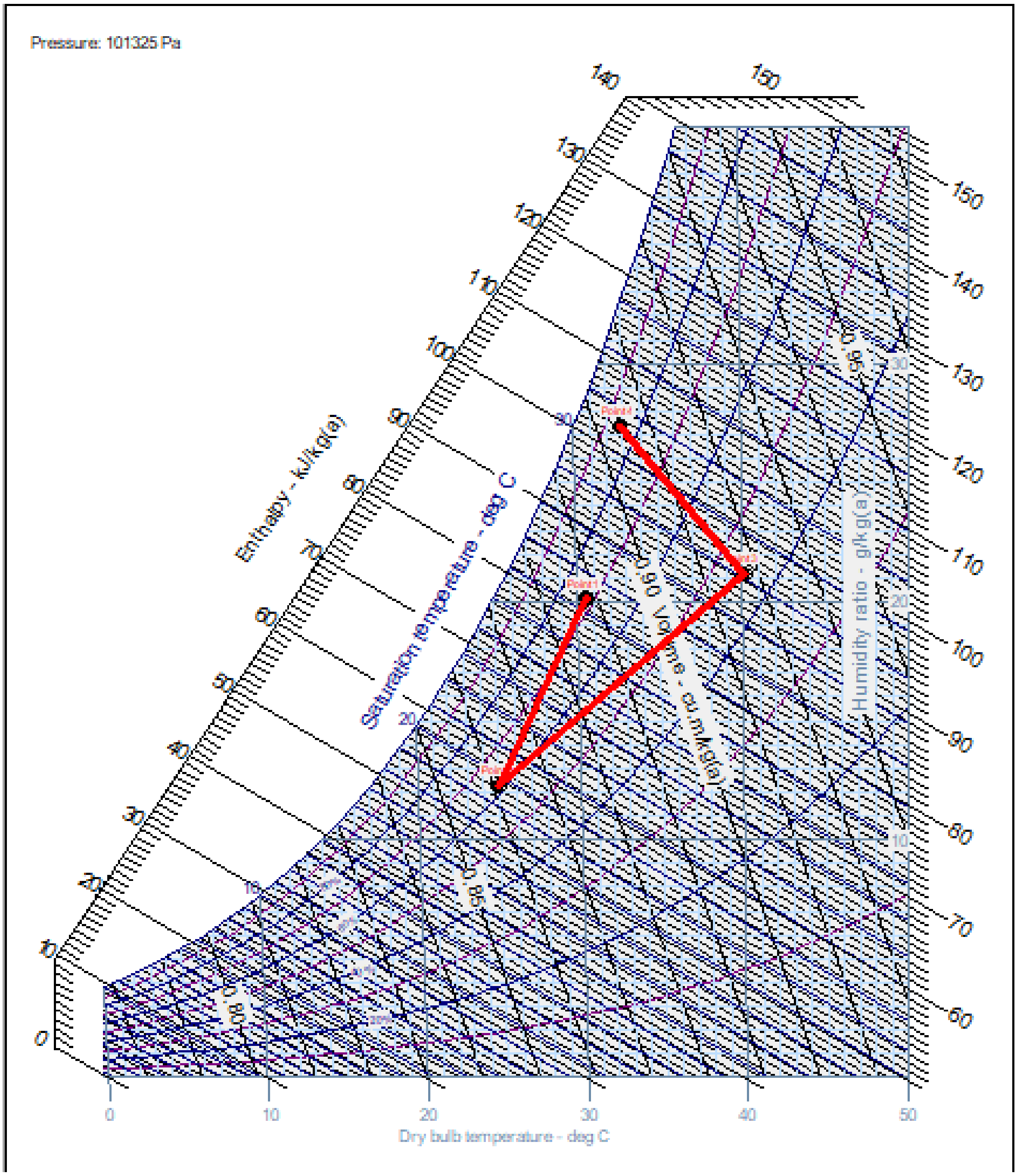

Figure 4: The process of the changing of airflow at v = 4.1 m/s for saturated steam

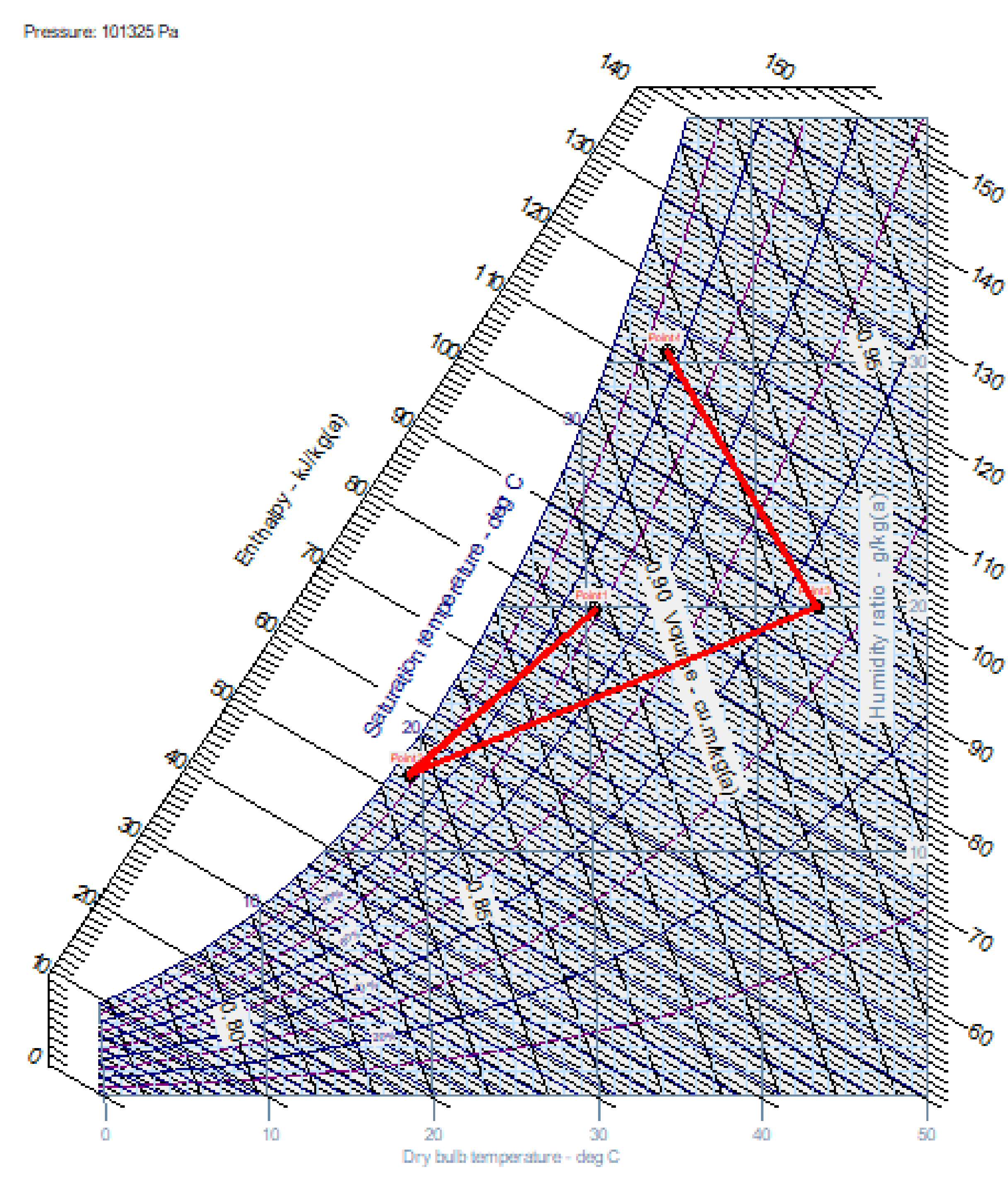

Figure 5: The process of the changing of airflow at v = 2.6 m/s for superheated steam

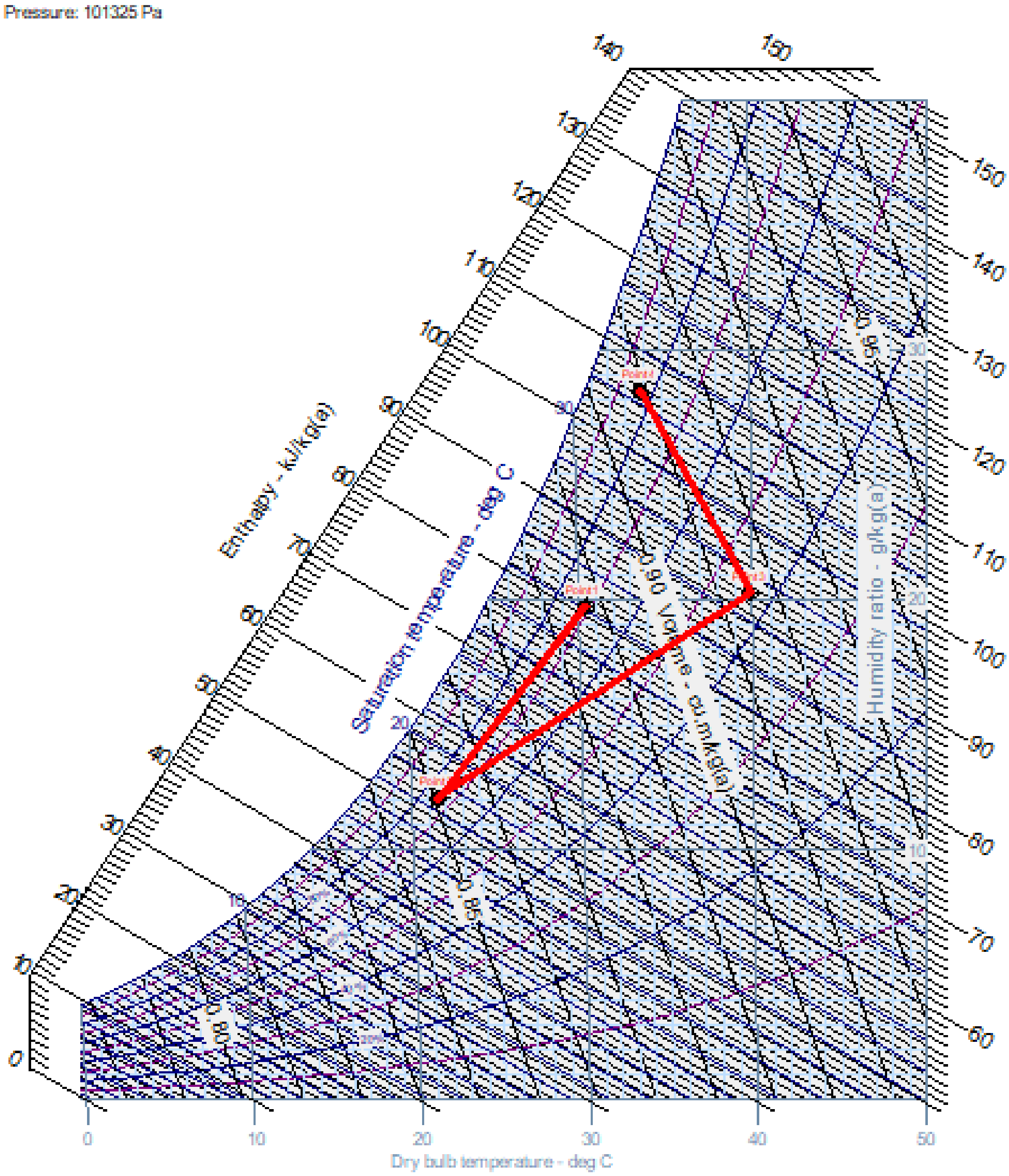

Figure 6: The process of the changing of airflow at v = 3.5 m/s for superheated steam

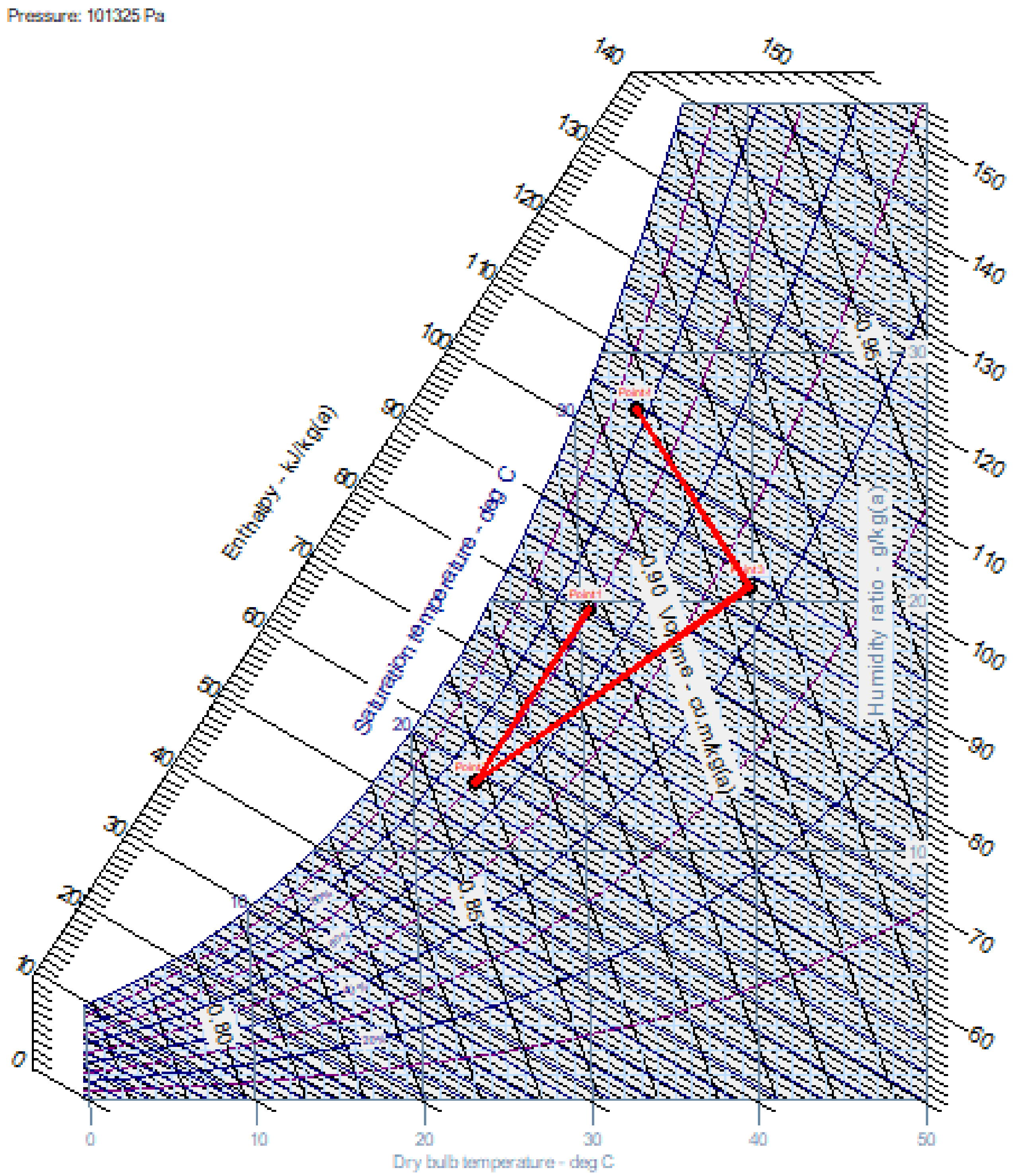

Figure 7: The process of the changing of airflow at v = 4.1 m/s for superheated steam

Table 4: Calculation

Saturated steam

v (m/s)	$t_{dry,1}$ (ºC)	ρ_1 (kg/m^3)	G_{kk} (kg/s)	Q_0 (kW)	G_{water} (kg/h)	G'_{water} (kg/h)	$t_{dry,3}$ (ºC)	ρ_3 (kg/m^3)	G'_{kk} (kg/s)	Q (kW)
2.6	30.25	1.1640	0.044	1.098	0.9413	1.080	44.10	1.1140	0.0417	1.7720
3.5	30.40	1.1634	0.059	2.375	2.0898	0.900	37.65	1.1360	0.0573	2.4113
4.1	30.55	1.1628	0.069	1.764	1.9277	0.438	40.10	1.1280	0.0666	2.5564

Superheated steam

v (m/s)	$t_{dry,1}$ (ºC)	ρ_1 (kg/m^3)	G_{kk} (kg/s)	Q_0 (kW)	G_{water} (kg/h)	G'_{water} (kg/h)	$t_{dry,3}$ (ºC)	ρ_3 (kg/m^3)	G'_{kk} (kg/s)	Q (kW)
2.6	30.50	1.1630	0.044	1.254	1.0659	0.522	43.55	1.1150	0.0418	1.7831
3.5	30.40	1.1634	0.059	1.683	1.6042	0.420	39.90	1.1280	0.0569	2.2747
4.1	30.50	1.1630	0.069	1.703	1.7056	0.315	39.80	1.1290	0.0666	2.4587

Note :

- *v*: velocity measured at the outlet of the gas pipe, m/s
- $t_{dry,1}$: dry temperature at the entrance of aerodynamic tube, ºC
- $t_{dry,3}$: dry temperature at the entrance of dryer, ºC
- ρ_1: specific mass of air before cooler, kg/m^3
- ρ_3: specific mass of air before dryer, kg/m^3
- G_{kk}: mass flowrate of air into the aerodynamic tube, kg/s
- G'_{kk}: mass flowrate of air before dryer, kg/s
- Q_0: cooling capacity of cooler, kJ/s
- G_{water}: flowrate of water condensed from the cooler in theory, kg/h.
- G'_{water}: flowrate of water condensed from the cooler in practice, kg/h
- Q: thermal load of air dryer, kW

6. DISCUSSION

6.1. Explain the change in air when traveling through the aerodynamic tube based on the change of the humidity of the air.

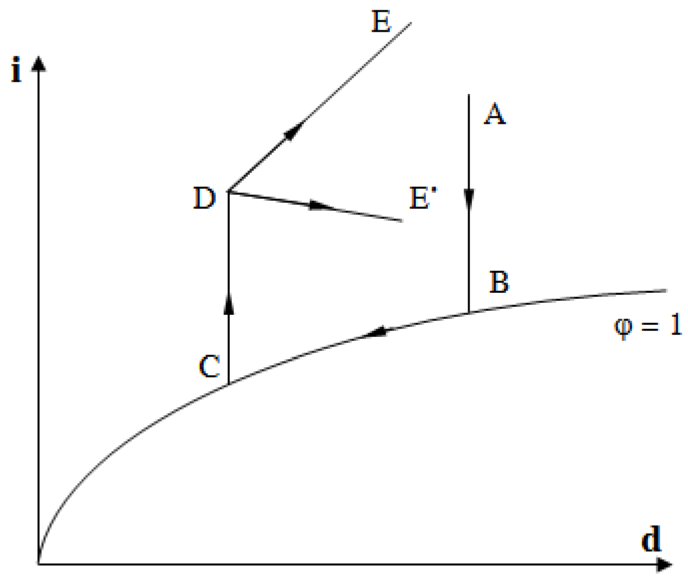

Figure 8: The diagram of the change in air when traveling through the aerodynamic tube in theory.

- *<u>Through the cooler:</u>* the change in the states of the air is represented by two straight lines AB and BC.

  - *In the early stage of the cooling process (AB)*: the absolute humidity d of the air remains unchanged (due to unchanged water content). Air temperature gradually decreases to the dewpoint temperature. Relative humidity φ increases up to saturated state φ = 1. At dewpoint temperature B, corresponding to saturated state, the water begins to condense.

  - *In the later stage of cooling (BC):* the relative humidity φ of the air remains unchanged and equals to 1, since the air is now saturated. As the air continues being cooled, the temperature of the air continues decreasing. Absolute humidity d of the air decreases as the condensed water makes the water content in humid air decrease.

- *<u>Through the dryer</u>*: the change in air state is represented by a straight line CD. The absolute humidity d of the air remains unchanged (due to unchanged water content) and the temperature of the air gradually increases. Relative humidity φ gradually decreases.

- *<u>Through the steam nozzle</u>*: the change in the states of the air is represented by a straight line:

- *If using saturated steam:* the change in the state of the air is represented by the DE. The absolute humidity d of the air increases as the air receives moisture. Enthalpy i also increases as the air receives more heat from saturated steam.

- *If using superheated steam:* the change in the state of the air is represented by a straight line between DE and DE'. The more superheated the water is, the closer that light is to DE'. The absolute humidity d of the air increases as the air receives moisture. Enthalpy i also increases the air receives heat from the superheated steam, but the increasing amount is smaller than using saturated steam.

6.2. Explain why you can determine the humidity of the air through the dry bulb and wet bulb temperature

- *Dry bulb temperature:* The temperature of the gas mixture determined by the normal thermometer.

  The dry bulb temperature is also the temperature of the air because the mercury directly contacts with the air.

- *Wet bulb temperature* is the steady temperature achieved when a very small amount of water evaporates into an unsaturated air mixture under adiabatic conditions.

The wet bulb temperature is measured by a conventional thermometer whose mercury bulb is covered with a wet cloth. Put water in the cup, when water evaporates adiabatically, it will take the heat around and make temperature of surrounding air decrease. Waiting until the temperature is stable, that temperature value is called wet bulb temperature. We have to keep the cup always filled with water.

The dryer the air is or the smaller its relative humidity φ is, the more water evaporates and the more heat surrounding air will be lost; therefore, the lower wet bulb temperature is (which means larger difference between wet bulb and ry bulb temperature). When the air is dry φ = 0, the difference between wet bulb and dry bulb temperature is the maximum. Otherwise, when the humid air is saturated φ = 100%, the water in the cup can not evaporate therefore wet bulb temperature is equal to dry bulb temperature or we say the difference between them is 0.

Wet bulb temperature is the saturated temperature corresponding to the saturated pressure of vapor in humid air.

In summary, the difference between dry bulb temperature and wet bulb temperature is characterized for the ability to get more moisture of the air. Technically, that difference is termed "drying potential ε". $\varepsilon = t_{dry} - t_{wet}$

- *Using i-d diagram to determine humidity of the air knowing wet bulb and dry bulb temperature:*

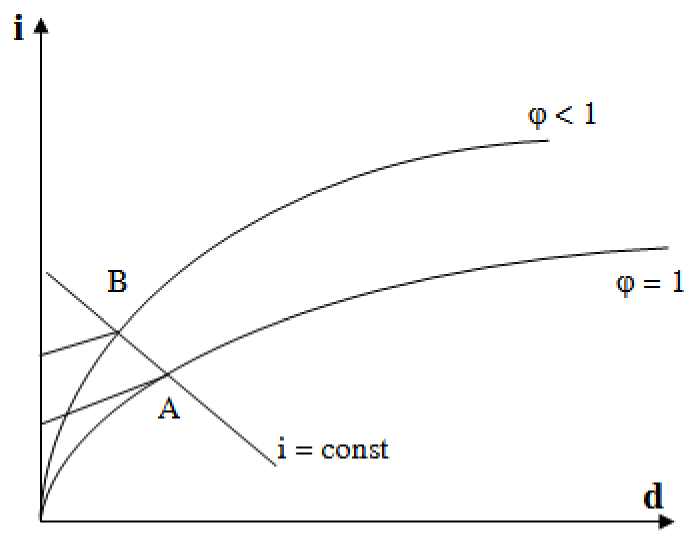

Figure 9: Way to determine humidity of the air, knowing wet bulb and dry bulb temperature.

t_{dry} t_{wet}

+ Draw a line through t_{wet}, parallel to t = const line, intersecs with φ = 1 at point A.
+ Draw a line through t_{dry}, parallel to t = const line
+ Draw a line through point A, parallel to i = const line, intersecs with t_{dry} line at point B.
+ B is the point indicating the state of air determined by t_{wet} and t_{dry} value.

- Relative humidity φ of air is determined using humidity meter. There are many types of humidity, each types works in different ways but still having the same basic thermodynamic principle as below:

+ Humidity meter determine the relative humidity of the air based on the difference between dry bulb and wet bulb temperature. Assume q_1 is the heat that the air supplies to the mercury bulb of wet bulb thermometer and q_2 is the heat that the water consumes to evaporate. We have:

$$q_1 = q_2 \quad (1)$$

According to heat transfer theory: $q_1 = \alpha\,(t_{dry} - t_{wet})$ (2)

$$q_2 = q_m.r \quad (3)$$

Where α: natural convection heat transfer coefficient, W / (m^2K).

r: latent heat, kJ / kg.

+ Evaporation intensity q_m [kg / (m^2s)] can be approximated by the Dalton formula through the evaporation coefficient α_m [kg / (m2.s.bar)] and the pressure difference between saturated pressure corresponding to wet bulb thermometer- p_m, and vapor pressure p_a in humid air.

$$q_m = \alpha_m\,(p_m - p_a)\,760 / B$$

Where B is the air ambient pressure where we determine the relative humidity φ.

+ If the air ambient pressure B is measured in bar, the above formula is written as:

$$q_m = \alpha_m (p_m - p_a)\frac{1{,}013}{B} \quad (4)$$

+ Substitute (2),(3),(4) into (1), we have:

$$p_m - p_a = \frac{\alpha}{\alpha_m.1{,}013.r} B(t_{dry} - t_{wet}) = A.B.(t_{dry} - t_{wet}) \quad (5)$$

with $A = \dfrac{\alpha}{\alpha_m.1{,}013.r}$

+ Factor A is called the humidity meter factor and depends on the heat transfer coefficient α and the evaporation coefficient α_m. These coefficients depend on the velocity of natural air motion. Thus, we can derive that A = f (*v*). Experimental results show that when the velocity $v < 0.5$ m/s, A = 66.10^{-5} and when $v > 0.5$ m/s, coefficient A is determined by the following formula:

$$A = \left(65 + \frac{6{,}75}{V}\right).10^{-5}$$

From (5) we can derive: $p_a = p_m - A.B.(t_{dry} - t_{wet})$

We also have: $\varphi = \dfrac{p_a}{p_b}$

⇨ We have a formula for determining the relative humidity φ of the air at saturated pressure p_b and the difference in temperature as follows:

$$\varphi = \frac{p_m}{p_b} - \frac{A.B}{p_b}(t - t_{ö}) \quad (6)$$

+ In equation (6), p_m and p_b are the saturation pressure corresponding to wet bulb temperature and dry bulb temperature, respectively.

+ So, it is quite possible to determine the relative humidity of the air by knowing the dry bulb temperature and wet bulb temperature.

6.3. Comparisons between the process cooling, drying and spraying to the wet air in diagram i-d of the theory and reality. Explain the difference.

- The change in the state of the air passing through the aerodynamic tube in practice is presented above in *Figure 2 - Figure 7*, but in general, they all have following trend:

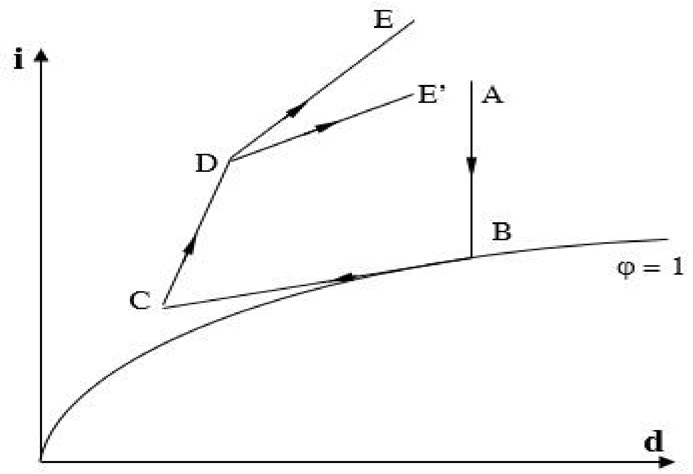

Figure 10: The diagram of the change in air when traveling through the aerodynamic tube in practice.

- *Through the cooler (lines AB, BC)*: the change of air state is not the same as the theory. At the end of cooling process (Point C), the air is not at saturated state as in theory but at an unsaturated state. This is because the air coming out of the cooler has received extra heat from the surroundings before reaching dry bulb and wet bulb thermometer.
- *Through the dryer (CD):* the change of air state is not the same as in theory. This drying process does not take place at a constant absolute humidity d condition as theoretically; in practice absolute humidity d gradually increases. This is because the air coming out of the dryer has received more moisture from the surroundings before reaching dry bulb and wet bulb thermometer.
- *Through the steam nozzel (DE):* the change of air state is the same as in theory because the surroundings does not significantly affect the results.

6.4. Comparisons between theoretical values and practical values

- *Capacity of cooler:* Theoretical capacity is larger than practical capacity because of the heat loss to the surroundings, so the cooler can not receive full heat load the air exposed. That’s why cooler can not perform the best capacity as in theory.
- *Condensed water from cooler*

- In theory: G_{water} = (0.9 ÷ 2.0) kg/h
- In practice: G'_{water} = (0.3 ÷ 1.0) kg/h

⇨ The difference between theory and practice maybe due to the error whens we take the condensed water sample and record the time.

- *Thermal load of Air dryer* : Practical thermal load is smaller than theoretical thermal load because of the heat loss to the surroundings, so the amount of heat the air received is smaller than amount of heat supplying by the dryer.

6.5. The cause of the errors between practice and theory.

a. The reliability of the above results is not high due to many factors affecting the experiment. Experiments are not as accurate as in theory. It should be done in tighter conditions, optimizing equipment to get the best results.

b. Causes of error:

+ *Errors when reading temperature values:* The temperature values shown on the screen are not stable. They usually vary up or down 1÷ 2°C so we can't get the accurate temperatures

+ *Errors when measuring the condensed water flowrate: 3 reasons:* We can not read accurately the value of the graduated cylinder; The self-error of the graduated cylinder; and The error when we record the time.

+ *Errors due to equipment:*self-error of equipment, the equipment is not well thermally and humidly insulated; the system is not closed so it will be affected by surroundings, : air flowrate into the equipment is not stable; equipment has just fixed so its work is not properly right;…

+ *Errors due to calculation*: when we looking up and rounding the data.

7. APPENDIX

7.1. *Determine parameters of air:*

- On i-d diagram, based on wet bulb and dry bulb temperature, we can determine Relative Humidity φ (%), Enthalpy I (kJ/kg) and Density d (kg/kg) of the air at different points
- Draw a line through t_{wet}, parallel to t = const line, intersecs with φ = 1 at point A.
- Draw a line through t_{dry}, parallel to t = const line
- Draw a line through point A, parallel to i = const line (this line shows us enthalpy I values at different states) , intersecs with t_{dry} line at point B.
- B is the point indicating the state of air determined by t_{wet} and t_{dry} value.
- φ = 0 line through B shows the Relative Humidity φ value of air

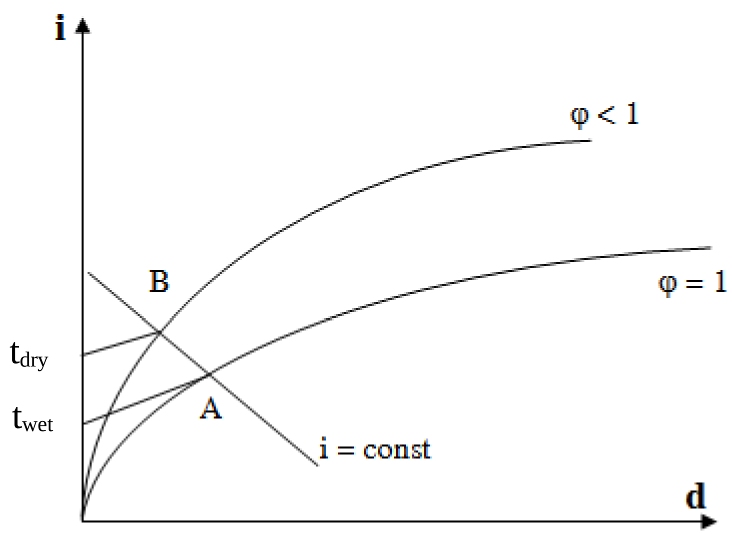

Figure 11: Way to determine parameters of air, knowing wet bulb and dry bulb temperature

7.2. Determine the air flowrate passing through the Aerodynamic tube

- Mass flowrate G_{kk} (kg/s) of air in aerodynamic tube can be determined using the following formula:

$$G_{kk} = v.F.\rho, \text{ kg/s} \quad (7)$$

Where:

+ v: velocity of the air measured at the outlet of aerodynamic tube (Table 2), m/s.
+ F = 0.0144 m^2: the outlet area of aerodynamic tube
+ ρ: specific mass of air (determined in Table 5), kg/m^3.
+ ρ is determined based on the dry bulb temperature (°C) at the point we want to examine

Table 5: The specific mass of air ρ (kg/m^3) depends on the temperature (°C) of it

t	30	31	32	33	34	35	36	37	38	39
ρ	1.165	1.161	1.157	1.154	1.15	1.146	1.142	1.139	1.135	1.131
t	40	41	42	43	44	45	46	47	48	49
ρ	1.128	1.124	1.121	1.117	1.114	1.11	1.107	1.103	1.1	1.096
t	50	51	52	53	54	55	56	57	58	59
ρ	1.093	1.089	1.086	1.083	1.079	1.076	1.073	1.07	1.066	1.063
t	60	61	62	63	64	65	66	67	68	69
ρ	1.06	1.057	1.054	1.051	1.047	1.044	1.041	1.039	1.035	1.032
t	70	71	72	73	74	75	76	77	78	79
ρ	1.029	1.026	1.023	1.02	1.017	1.014	1.011	1.009	1.006	1.003

7.3. Calculation for cooler

a. *Cooling capacity of cooler Q_0*

$$Q_0 = G_{kk}\,(i_1 - i_2), \text{ kW}$$

Where:

+ G_{kk}: mass flowrate of air in aerodynamic tube, determined by the formula (7), kg/s.
+ i_1, i_2: enthalpy of air before and after the cooler (Table 3), kJ/kg.

b. The theoretical volume of water condensed from the cooler G_{water}

$$G_{water} = 3600.G_{kk}.(d_2 - d_1), \text{ kg/hour}$$

Where:

+ d_1, d_2: density of the air before and after the cooler (Table 3), kg/kg.

c. The practical volume of water condensed from the cooler G'_{water}

$$G'_{water} = \frac{0.06\,.\,V_1}{t_1}, \text{ kg/hour} \quad (8)$$

Where:

+ V_1: volume of water condensed from the cooler (Table 3), ml.
+ t_1: amount time taken to get water sample (Table 3), minutes.

Note: As we do the experiment twice to get more accurate result, we have to calculate the average flowrate from 2 experiments, then we substitute it to the equation to calculate G'_{water}

7.4. Calculation for Air dryer

a. Heat load Q of Air Dryer

$$Q = G'_{kk} (i_3 - i_2) \text{ , kW}$$

Where:

+ i_2, i_3: enthalpy of air before and after the air dryer (Table 3), kJ/kg.

b. The amount of heat generated by the power supply through the resistor.

$$Q' = 1\text{kW (one resistor)}$$
$$Q' = 2\text{kW (two resistors)}$$

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